

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>MONOLITH — Orbital Architecture</title>

<meta name="description" content="We design and engineer the habitats and infrastructure for
humanity's next chapter in space.">

<script src="https://cdn.tailwindcss.com"></script>

<script src="https://unpkg.com/lucide@latest"></script>

<script src="https://cdn.jsdelivr.net/npm/three.js/r128/three.min.js"></script>

<script src="https://cdn.jsdelivr.net/npm/gsap@3.12.5/gsap.min.js"></script>

<script src="https://cdn.jsdelivr.net/npm/gsap@3.12.5/ScrollTrigger.min.js"></script>

<link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800;900&family=
DM+Serif+Display:ital@0;1&display=swap" rel="stylesheet">

<script>

tailwind.config={theme:{extend:{fontFamily:{sans:['Inter','sans-serif'],serif:['DM Serif
Display','serif']}}}}

</script>

<style>

*{margin:0;padding:0;box-sizing:border-box}

html{scroll-behavior:smooth}

body{font-family:'Inter',sans-serif;background:#060608;color:#d0ccc8;overflow-x:hidden;-webkit-
font-smoothing:antialiased}

::-webkit-scrollbar{width:3px}::-webkit-scrollbar-track{background:#060608}::-webkit-scrollbar-
thumb{background:#1a1816;border-radius:2px}

::selection{background:rgba(212,98,59,.12);color:#fff}

/* Grid */

.grid-ov{position:fixed;inset:0:pointer-events:none;z-index:2;

background-image:linear-gradient(rgba(208,204,200,.01) 1px,transparent 1px),linear-
gradient(90deg,rgba(208,204,200,.01) 1px,transparent 1px);
```

```
background-size:calc(100%/12) 80px}
```

```
/* Noise */
```

```
body::after{content:'';position:fixed;inset:0;background-image:url("data:image/svg+xml,%3Csvg  
viewBox='0 0 512 512' xmlns='http://www.w3.org/2000/svg'%3E%3Cfilter  
id='n'%3E%3CfeTurbulence type='fractalNoise' baseFrequency='.8' numOctaves='4'  
stitchTiles='stitch'/%3E%3C/filter%3E%3Crect width='100%25' height='100%25' filter='url(%23n)'  
opacity='.018'/%3E%3C/svg%3E");pointer-events:none;z-index:9990;opacity:.5}
```

```
/* Loader */
```

```
#loader{position:fixed;inset:0;z-index:10000;background:#060608;display:flex;align-  
items:center;justify-content:center;flex-direction:column;transition:opacity .6s,visibility .6s}
```

```
#loader.out{opacity:0;visibility:hidden;pointer-events:none}
```

```
.ld-bar{width:120px;height:1px;background:#141210;overflow:hidden}
```

```
.ld-fill{height:100%;width:0;background:#d0ccc8;transition:width .2s}
```

```
.ld-num{margin-top:10px;font-size:10px;letter-spacing:.15em;color:#28241f;font-variant-  
numeric:tabular-nums}
```

```
/* Canvas */
```

```
#c3d{position:fixed;inset:0;z-index:1;pointer-events:none}
```

```
/* Sidebar */
```

```
#sidebar{position:fixed;left:0;top:50%;transform:translateY(-50%);z-index:40;width:175px;padding:0  
16px}
```

```
.sb-link{display:flex;align-items:center;gap:7px;padding:3px 0;cursor:pointer;transition:color .3s}
```

```
.sb-d{width:8px;height:1px;background:#1a1816;transition:all .3s}
```

```
.sb-t{font-size:8.5px;letter-spacing:.1em;text-transform:uppercase;color:#28241f;transition:color .3s}
```

```
.sb-link.active .sb-d{width:16px;background:#D4623B}
```

```
.sb-link.active .sb-t{color:#d0ccc8}
```

```
.sb-link:hover .sb-t{color:#504a44}
```

```
/* Nav */
```

```
#topnav{position:fixed;top:0;left:0;right:0;z-index:50;transition:all .35s cubic-bezier(.16,1,.3,1)}
```

```
#topnav.scrolled{background:rgba(6,6,8,.9);backdrop-filter:blur(18px);-webkit-backdrop-filter:blur(18px);border-bottom:1px solid rgba(208,204,200,.025)}
```

```
#topnav.hide{transform:translateY(-100%)}
```

```
.nv-l{position:relative;font-size:9.5px;letter-spacing:.13em;text-transform:uppercase;color:#504a44;transition:color .3s}
```

```
.nv-l:hover{color:#d0ccc8}
```

```
.nv-l::after{content:"";position:absolute;bottom:-2px;left:0;width:100%;height:1px;background:#d0ccc8;transform:scaleX(0);transform-origin:right;transition:transform .3s cubic-bezier(.16,1,.3,1)}
```

```
.nv-l:hover::after{transform:scaleX(1);transform-origin:left}
```

```
/* Hero */
```

```
.hl{overflow:hidden;display:block}.hl span{display:inline-block;transform:translateY(108%);transition:transform .85s cubic-bezier(.16,1,.3,1)}.hl span.in{transform:translateY(0)}
```

```
/* Reveal */
```

```
.rv{opacity:0;transform:translateY(30px);transition:opacity .7s cubic-bezier(.16,1,.3,1),transform .7s cubic-bezier(.16,1,.3,1)}
```

```
.rv.in{opacity:1;transform:translateY(0)}
```

```
.rv-d1{transition-delay:.06s}.rv-d2{transition-delay:.12s}.rv-d3{transition-delay:.18s}.rv-d4{transition-delay:.24s}.rv-d5{transition-delay:.3s}
```

```
/* Watermark */
```

```
.wn{position:absolute;top:-6px;right:14px;font-size:clamp(4.5rem,10vw,9rem);font-weight:800;line-height:.85;color:rgba(208,204,200,.01);pointer-events:none;user-select:none;z-index:0}
```

```
/* Service row */
```

```
.svc{border-bottom:1px solid rgba(208,204,200,.03);padding:1.6rem 0;display:grid;grid-template-columns:40px 1fr auto;gap:1.6rem;align-items:center;transition:all .3s;cursor:default}
```

```
.svc:hover{padding-left:.7rem}
```

```
.svc:hover .svc-n{color:#D4623B}
```

```
.svc:hover .svc-a{opacity:1;transform:translateX(0)}
```

```
.svc-n{font-variant-numeric:tabular-nums;color:#1a1816;font-size:11px;transition:color .3s}
```

```
.svc-a{opacity:0;transform:translateX(-4px);transition:all .3s}
```

```
/* Demo card */
```

```
.dem{position:relative;overflow:hidden;border:1px solid rgba(208,204,200,.03);transition:border-color .3s;cursor:pointer}
```

```
.dem:hover{border-color:rgba(208,204,200,.08)}
```

```
.dem img{transition:transform .7s cubic-bezier(.16,1,.3,1)}
```

```
.dem:hover img{transform:scale(1.02)}
```

```
.dem .dem-c{opacity:0;transform:translateY(5px);transition:all .3s}
```

```
.dem:hover .dem-c{opacity:1;transform:translateY(0)}
```

```
/* Process */
```

```
.prc{position:relative;padding-left:2rem;padding-bottom:2rem}
```

```
.prc::before{content:"";position:absolute;left:0;top:4px;width:1px;height:calc(100% - 4px);background:rgba(208,204,200,.03)}
```

```
.prc:last-child::before{display:none}
```

```
.prc-d{width:4px;height:4px;border-radius:50%;background:#1a1816;position:absolute;left:-2px;top:3px;transition:background .3s}
```

```
.prc:hover .prc-d{background:#D4623B}
```

```
/* FAQ */
```

```
.fq-i{border-bottom:1px solid rgba(208,204,200,.03)}
```

```
.fq-q{padding:1.1rem 0;cursor:pointer;display:flex;justify-content:space-between;align-items:center;gap:2rem}
```

```
.fq-a{max-height:0;overflow:hidden;transition:max-height .4s cubic-bezier(.16,1,.3,1),padding .3s}
```

```
.fq-a.open{max-height:160px;padding-bottom:1.1rem}
```

```
.fq-ic{transition:transform .3s}
```

```
/* Activity */
```

```
.ac{width:8px;height:8px;border-radius:1.5px;background:#0c0b0a;transition:background .2s}
```

```
.ac.l1{background:#1f1610}.ac.l2{background:#2d1e0f}.ac.l3{background:#3d2810}.ac.l4{background:#4d3212}.ac.l5{background:#D4623B}
```

```
/* Pipeline grid item */
```

```
.pl-item{border-bottom:1px solid rgba(208,204,200,.03);padding:1.8rem 0;display:grid;grid-template-columns:140px 1fr;gap:1.5rem;align-items:start}
```

```
/* Mobile menu */
```

```
#mobmenu{position:fixed;inset:0;z-index:45;background:#060608;transform:translateY(-100%);transition:transform .5s cubic-bezier(.16,1,.3,1);display:flex;flex-direction:column;justify-content:center;padding:3rem 2rem}
```

```
#mobmenu.open{transform:translateY(0)}
```

```
/* Cookie */
```

```
#cookie{position:fixed;bottom:0;left:0;right:0;z-index:200;transform:translateY(120%);transition:transform .5s cubic-bezier(.16,1,.3,1)}
```

```
#cookie.show{transform:translateY(0)}
```

```
/* Scene ind */
```

```
#sc-ind{position:fixed;bottom:18px;left:50%;transform:translateX(-50%);z-index:30;font-size:8.5px;letter-spacing:.16em;text-transform:uppercase;color:#1a1816;transition:color .3s;pointer-events:none}
```

```
/* Toast */
```

```
#toast{position:fixed;bottom:18px;right:18px;z-index:300;background:#0e0d0c;border:1px solid rgba(208,204,200,.04);padding:10px 16px;border-radius:4px;display:flex;align-items:center;gap:7px;opacity:0;transform:translateY(8px);transition:all .4s cubic-bezier(.16,1,.3,1);pointer-events:none}
```

```
#toast.show{opacity:1;transform:translateY(0);pointer-events:auto}
```

```
/* HR anim */
```

```
.hr-a{transform:scaleX(0);transform-origin:left;transition:transform .6s cubic-bezier(.16,1,.3,1)}
```

```
.hr-a.in{transform:scaleX(1)}
```

```
@media(max-width:1024px){#sidebar{display:none}#sc-ind{display:none}}
```

```
@media(max-width:768px){
```

```
.svc{grid-template-columns:30px 1fr auto;gap:.6rem;padding:1rem 0}
```

```

.svc:hover{padding-left:.3rem}

.pl-item{grid-template-columns:1fr;gap:.4rem}
}

</style>

</head>

<body>

<div id="loader"><div class="ld-bar"><div class="ld-fill" id="ld-f"></div></div><div class="ld-num"
id="ld-n">000</div></div>

<div class="grid-ov"></div>

<canvas id="c3d"></canvas>

<div id="sc-ind">SCENE — <span id="sc-name">ORBIT</span></div>

<!-- SIDEBAR -->

<aside id="sidebar">

  <div class="mb-4" style="font-size:7.5px;letter-spacing:.16em;text-
transform:uppercase;color:#1a1816">On this page</div>

  <div class="space-y-[1px]" id="sb-links">

    <div class="sb-link active" data-target="start"><span class="sb-d"></span><span class="sb-
t">Start</span></div>

    <div class="sb-link" data-target="numbers"><span class="sb-d"></span><span class="sb-t">By the
Numbers</span></div>

    <div class="sb-link" data-target="build"><span class="sb-d"></span><span class="sb-t">What we
build</span></div>

    <div class="sb-link" data-target="engineer"><span class="sb-d"></span><span class="sb-t">How
we engineer</span></div>

    <div class="sb-link" data-target="work"><span class="sb-d"></span><span class="sb-
t">Work</span></div>

    <div class="sb-link" data-target="studio"><span class="sb-d"></span><span class="sb-
t">Studio</span></div>

    <div class="sb-link" data-target="process"><span class="sb-d"></span><span class="sb-
t">Process</span></div>

    <div class="sb-link" data-target="faq"><span class="sb-d"></span><span class="sb-
t">FAQ</span></div>

```

```
<div class="sb-link" data-target="architecture"><span class="sb-d"></span><span class="sb-t">The
Architecture</span></div>
```

```
<div class="sb-link" data-target="activity"><span class="sb-d"></span><span class="sb-
t">Activity</span></div>
```

```
<div class="sb-link" data-target="contact"><span class="sb-d"></span><span class="sb-
t">Contact</span></div>
```

```
<div class="sb-link" data-target="footer"><span class="sb-d"></span><span class="sb-
t">Footer</span></div>
```

```
</div>
```

```
<div class="mt-5 pt-4 border-t" style="border-color:rgba(208,204,200,.03)">
```

```
<div style="font-size:7.5px;letter-spacing:.16em;text-transform:uppercase;color:#1a1816"
class="mb-2">Navigation</div>
```

```
<div class="space-y-1">
```

```
<div class="flex items-center gap-1.5"><span class="text-[#28241f] text-[9px]">↑
↓</span><span class="text-[#28241f] text-[9px]">W S</span><span class="text-[#28241f] text-[8px]
tracking-wider">Change scene</span></div>
```

```
<div class="flex items-center gap-1.5"><span class="text-[#28241f] text-[9px]">K</span><span
class="text-[#28241f] text-[8px] tracking-wider">Contact</span></div>
```

```
</div>
```

```
</div>
```

```
<div class="mt-3"><a href="#contact" class="text-[#28241f] hover:text-[#D4623B] transition-colors"
style="font-size:8px;letter-spacing:.13em;text-transform:uppercase">Contact ↗</a></div>
```

```
</aside>
```

```
<!-- TOP NAV -->
```

```
<nav id="topnav">
```

```
<div class="max-w-[1400px] mx-auto px-5 lg:px-9 flex items-center justify-between h-[60px] lg:h-
[68px]">
```

```
<a href="#start" class="text-[12px] font-bold tracking-[-.03em]">MONOLITH</a>
```

```
<div class="hidden md:flex items-center gap-7 lg:gap-8">
```

```
<a href="#build" class="nv-l">Services</a>
```

```
<a href="#studio" class="nv-l">Studio</a>
```

```
<a href="#process" class="nv-l">Process</a>
```

```
<a href="#contact" class="nv-l">Contact</a>
```

</div>

Start a project

<button id="menubtn" class="md:hidden w-6 h-6 flex flex-col items-end justify-center gap-[3.5px]">

</button>

</div>

</nav>

<!-- MOBILE MENU -->

<div id="mobmenu">

Start

What we build

Work

Process

Contact

<div class="absolute bottom-8 left-8 text-[#28241f] text-[9px] tracking-wider">ops@monolith.space</div>

</div>

<main class="relative z-10">

<!-- START -->

<section id="start" class="relative min-h-screen flex items-end pb-12 md:pb-20 overflow-hidden">


```
<div class="absolute inset-0 bg-gradient-to-b from-transparent via-transparent to-[#060608] z-[3]
pointer-events-none"></div>

<div class="relative z-[4] w-full max-w-[1400px] mx-auto px-5 lg:px-9 lg:pl-[195px]">

  <div class="mb-4 rv" id="h-tag"><span class="text-[9.5px] tracking-[.16em] uppercase text-
[#504a44]">00 — Orbital Architecture · Habitat Design · Surface Systems</span></div>

  <div class="max-w-[600px]">

    <h1 class="text-[clamp(2.4rem,6.2vw,6.2rem)] font-bold leading-[.87] tracking-[-.04em] mb-6">

      <span class="hl"><span>MONO</span></span><span class="hl"><span class="font-serif italic
font-normal text-[#504a44]">LITH</span></span>

    </h1>

    <p class="text-[#504a44] text-[13px] leading-[1.65] max-w-[380px] mb-8 rv rv-d2" id="h-sub">We
design and engineer the habitats and infrastructure for humanity's next chapter in space.</p>

    <div class="flex items-center gap-3.5 rv rv-d3" id="h-cta">

      <a href="#numbers" class="group flex items-center gap-2">

        <span class="w-7 h-7 rounded-full border flex items-center justify-center group-hover:border-
[rgba(208,204,200,.15)] transition-colors" style="border-color:rgba(208,204,200,.06)"><i data-
lucide="arrow-down" class="w-3 h-3 text-[#504a44]"></i></span>

        <span class="text-[9px] tracking-[.13em] uppercase text-[#504a44] group-hover:text-[#504a44]
transition-colors">Scroll to explore</span>

      </a>

    </div>

  </div>

</div>

</section>
```

```
<!-- BY THE NUMBERS -->
```

```
<section id="numbers" class="relative py-16 md:py-28 border-t" style="border-
color:rgba(208,204,200,.03)">
```

```
<div class="wn">01</div>
```

```
<div class="max-w-[1400px] mx-auto px-5 lg:px-9 lg:pl-[195px]">
```

```
<span class="text-[9.5px] tracking-[.16em] uppercase text-[#28241f] block mb-3.5 rv">01 — By the
Numbers</span>
```

```
<div class="hr-a h-px mb-8 rv rv-d1" style="background:rgba(208,204,200,.04)"></div>
```

<h2 class="text-[clamp(1.5rem,2.8rem,2.5rem)] font-bold tracking-[-.03em] leading-[-.95] mb-3 rv rv-d1">By the Numbers</h2>

<p class="text-[#504a44] text-[12px] max-w-[340px] mb-12 rv rv-d2">What the work of recent programmes measurably left behind.</p>

<div class="grid grid-cols-2 md:grid-cols-4 gap-7 md:gap-9">

<div class="rv rv-d1"><div class="text-4xl md:text-[3.2rem] font-bold tracking-[-.04em] mb-1 sv" data-t="14" data-s="">0</div><div class="text-[9px] tracking-[-.11em] uppercase text-[#28241f]">Habitat modules</div></div>

<div class="rv rv-d2"><div class="text-4xl md:text-[3.2rem] font-bold tracking-[-.04em] mb-1 sv" data-t="8" data-s="k">0</div><div class="text-[9px] tracking-[-.11em] uppercase text-[#28241f]">m² living space</div></div>

<div class="rv rv-d3"><div class="text-4xl md:text-[3.2rem] font-bold tracking-[-.04em] mb-1 sv" data-t="6" data-s="">0</div><div class="text-[9px] tracking-[-.11em] uppercase text-[#28241f]">Programmes delivered</div></div>

<div class="rv rv-d4"><div class="text-4xl md:text-[3.2rem] font-bold tracking-[-.04em] mb-1 sv" data-t="3" data-s="">0</div><div class="text-[9px] tracking-[-.11em] uppercase text-[#28241f]">Agency partners</div></div>

</div>

</div>

</section>

<!-- WHAT WE BUILD -->

<section id="build" class="relative py-16 md:py-28 border-t" style="border-color:rgba(208,204,200,.03)">

<div class="wn">02</div>

<div class="max-w-[1400px] mx-auto px-5 lg:px-9 lg:pl-[195px]">

02 — What we build

<div class="hr-a h-px mb-8 rv rv-d1" style="background:rgba(208,204,200,.04)"></div>

<h2 class="text-[clamp(1.5rem,2.8rem,2.5rem)] font-bold tracking-[-.03em] leading-[-.95] mb-3 rv rv-d1">What we build</h2>

<p class="text-[#504a44] text-[12px] max-w-[340px] mb-10 rv rv-d2">From a single-module demonstrator to a full station architecture — structures designed for the void.</p>

<div class="max-w-[700px] rv rv-d3">

<div class="svc">01<div><h3 class="text-[14px] font-semibold tracking-tight mb-0.5">Habitat Modules</h3><p class="text-[#504a44] text-[12px] hidden md:block">Pressurised living and working modules — structural analysis, interior layout, ECLSS

integration, and radiation shielding.</p></div><i data-lucide="arrow-right" class="svc-a w-3.5 h-3.5 text-[#504a44]"></i></div>

<div class="svc">02<div><h3 class="text-[14px] font-semibold tracking-tight mb-0.5">Surface Infrastructure</h3><p class="text-[#504a44] text-[12px] hidden md:block">Lunar and Martian surface systems — habitation cores, airlocks, dust mitigation, and thermal management for extreme environments.</p></div><i data-lucide="arrow-right" class="svc-a w-3.5 h-3.5 text-[#504a44]"></i></div>

<div class="svc">03<div><h3 class="text-[14px] font-semibold tracking-tight mb-0.5">Orbital Stations</h3><p class="text-[#504a44] text-[12px] hidden md:block">Complete station architectures — truss systems, solar arrays, docking interfaces, and orbital debris assessment.</p></div><i data-lucide="arrow-right" class="svc-a w-3.5 h-3.5 text-[#504a44]"></i></div>

</div>

</div>

</section>

<!-- HOW WE ENGINEER -->

<section id="engineer" class="relative py-16 md:py-28 border-t" style="border-color:rgba(208,204,200,.03)">

<div class="wn">03</div>

<div class="max-w-[1400px] mx-auto px-5 lg:px-9 lg:pl-[195px]">

03 — How we engineer

<div class="hr-a h-px mb-8 rv rv-d1" style="background:rgba(208,204,200,.04)"></div>

<h2 class="text-[clamp(1.5rem,2.8rem,2.5rem)] font-bold tracking-[-.03em] leading-[.95] mb-3 rv rv-d1">And how we engineer</h2>

<p class="text-[#504a44] text-[12px] max-w-[340px] mb-10 rv rv-d2">Design is one half. The other half is making it survivable.</p>

<div class="max-w-[700px] rv rv-d3">

<div class="svc">04<div><h3 class="text-[14px] font-semibold tracking-tight mb-0.5">Structural Analysis</h3><p class="text-[#504a44] text-[12px] hidden md:block">Finite element analysis under launch loads, pressurisation, thermal cycling, and micrometeoroid impact scenarios.</p></div><i data-lucide="arrow-right" class="svc-a w-3.5 h-3.5 text-[#504a44]"></i></div>

<div class="svc">05<div><h3 class="text-[14px] font-semibold tracking-tight mb-0.5">Thermal & ECLSS</h3><p class="text-[#504a44] text-[12px] hidden md:block">Thermal modelling, radiator sizing, and environmental control — air, water, and waste

recycling for closed-loop life support.</p></div><i data-lucide="arrow-right" class="svc-a w-3.5 h-3.5 text-[#504a44]"></i></div>

<div class="svc">06<div><h3 class="text-[14px] font-semibold tracking-tight mb-0.5">Digital Twin & Sim</h3><p class="text-[#504a44] text-[12px] hidden md:block">Real-time simulation of the complete habitat — system behaviour, failure modes, crew operations, and long-duration performance.</p></div><i data-lucide="arrow-right" class="svc-a w-3.5 h-3.5 text-[#504a44]"></i></div>

</div>

</div>

</section>

<!-- WORK -->

<section id="work" class="relative py-16 md:py-28 border-t" style="border-color:rgba(208,204,200,.03);background:rgba(6,6,8,.96)">

<div class="wn">04</div>

<div class="max-w-[1400px] mx-auto px-5 lg:px-9 lg:pl-[195px]">

04 — Work

<div class="hr-a h-px mb-8 rv rv-d1" style="background:rgba(208,204,200,.04)"></div>

<h2 class="text-[clamp(1.5rem,2.8rem,2.5rem)] font-bold tracking-[-.03em] leading-[-.95] mb-3 rv rv-d1">Work that holds pressure</h2>

<p class="text-[#504a44] text-[12px] max-w-[340px] mb-12 rv rv-d2">Three habitat programmes — concept through to engineering review.</p>

<div class="dem mb-4 rv rv-d2">

<div class="aspect-[2.1/1] md:aspect-[21/8] relative overflow-hidden">

<div class="absolute inset-0 bg-gradient-to-t from-black/75 via-black/10 to-transparent"></div>

<div class="absolute top-3.5 left-3.5">Programme 01 · Lunar habitat</div>

<div class="absolute bottom-0 left-0 right-0 p-5 md:p-7">

<div class="text-[#504a44] text-[9.5px] tracking-[-.13em] uppercase mb-1">Ares Base</div>

<h3 class="text-lg md:text-xl font-bold tracking-tight mb-1">A 6-module lunar surface habitat for a 12-crew research outpost — integrated ECLSS, dust-shielded airlocks, and in-situ resource utilisation.</h3>

View programme <i data-lucide="arrow-right" class="w-2.5 h-2.5"></i>

</div>

</div>

</div>

<div class="grid grid-cols-1 md:grid-cols-2 gap-4">

<div class="dem rv rv-d3">

<div class="aspect-[4/3] relative overflow-hidden">

<div class="absolute inset-0 bg-gradient-to-t from-black/75 via-black/10 to-transparent"></div>

<div class="absolute top-3.5 left-3.5">Programme 02 · LEO station</div>

<div class="absolute bottom-0 left-0 right-0 p-5">

<div class="text-[#504a44] text-[9.5px] tracking-[.13em] uppercase mb-1">Hermes Node</div>

<h3 class="text-base font-bold tracking-tight mb-0.5">An 8-module LEO logistics hub with automated docking, 30kW power bus, and crew rotation capability for 16.</h3>

View programme <i data-lucide="arrow-right" class="w-2.5 h-2.5"></i>

</div>

</div>

</div>

<div class="dem rv rv-d4">

<div class="aspect-[4/3] relative overflow-hidden">

<div class="absolute inset-0 bg-gradient-to-t from-black/75 via-black/10 to-transparent"></div>

<div class="absolute top-3.5 left-3.5">Programme 03 · Deep space</div>

```
<div class="absolute bottom-0 left-0 right-0 p-5">

  <div class="text-[#504a44] text-[9.5px] tracking-[.13em] uppercase mb-1">Olympus</div>

  <h3 class="text-base font-bold tracking-tight mb-0.5">A centrifugal gravity habitat for Mars
transit — 40m diameter, 0.38g artificial gravity, 24-month mission duration.</h3>

  <span class="dem-c inline-flex items-center gap-1.5 text-[9px] tracking-[.13em] uppercase
text-[#D4623B]">View programme <i data-lucide="arrow-right" class="w-2.5 h-2.5"></i></span>

</div>

</div>

</div>

</div>

</div>

</div>

</section>
```

```
<!-- STUDIO -->

<section id="studio" class="relative py-16 md:py-28 border-t" style="border-
color:rgba(208,204,200,.03)">

  <div class="wn">05</div>

  <div class="max-w-[1400px] mx-auto px-5 lg:px-9 lg:pl-[195px]">

    <span class="text-[9.5px] tracking-[.16em] uppercase text-[#28241f] block mb-3.5 rv">05 —
Studio</span>

    <div class="hr-a h-px mb-8 rv rv-d1" style="background:rgba(208,204,200,.04)"></div>

    <div class="grid grid-cols-1 lg:grid-cols-12 gap-10 lg:gap-6 items-center">

      <div class="lg:col-span-5">

        <h2 class="text-[clamp(1.5rem,2.8rem,2.5rem)] font-bold tracking-[-.03em] leading-[.95] mb-5
rv rv-d1">All under <span class="font-serif italic font-normal text-[#504a44]">one roof</span></h2>

        <div class="space-y-3.5 text-[#504a44] text-[12px] leading-[1.65] rv rv-d2">

          <p>Monolith is an orbital architecture practice. We don't do buildings on Earth — we do
structures in space. This focus means every engineer understands pressurisation, radiation, and zero-
g behaviour.</p>

          <p>Structural, thermal, ECLSS, and digital twin — all from one team. No handoffs, no gaps.</p>

        </div>

        <div class="flex flex-wrap gap-1.5 mt-7 rv rv-d3">

          <span class="text-[8.5px] tracking-[.12em] uppercase border px-2.5 py-1 rounded-full text-
[#28241f]" style="border-color:rgba(208,204,200,.04)">ANSYS</span>
```

SolidWorks

Siemens NX

MATLAB

</div>

</div>

<div class="lg:col-span-5 lg:col-start-7 rv rv-d2">

<div class="relative">

<div class="absolute -bottom-3.5 -left-3.5 bg-[#0a0908] border px-4 py-3" style="border-color:rgba(208,204,200,.04)">

<div class="text-lg font-bold tracking-[-.03em]">Munich / Remote</div>

<div class="text-[8.5px] text-[#28241f] tracking-[.13em] uppercase mt-0.5">Studio location</div>

</div>

</div>

</div>

</div>

</div>

</section>

<!-- PROCESS -->

<section id="process" class="relative py-16 md:py-28 border-t" style="border-color:rgba(208,204,200,.03)">

<div class="wn">06</div>

<div class="max-w-[1400px] mx-auto px-5 lg:px-9 lg:pl-[195px]">

06 — Process

<div class="hr-a h-px mb-8 rv rv-d1" style="background:rgba(208,204,200,.04)"></div>

<h2 class="text-[clamp(1.5rem,2.8rem,2.5rem)] font-bold tracking-[-.03em] leading-[.95] mb-3 rv rv-d1">From brief to review</h2>

<p class="text-[#504a44] text-[12px] max-w-[340px] mb-12 rv rv-d2">Four phases. No surprises.</p>

<div class="max-w-[520px] space-y-0">

<div class="prc rv rv-d1"><div class="prc-d"></div><div class="text-[#28241f] text-[10px] font-mono mb-1">01</div><h3 class="text-[14px] font-semibold tracking-tight mb-0.5">Concept</h3><p class="text-[#504a44] text-[12px] leading-[1.6]">Define mission requirements, crew size, duration, and mass budget. Establish the module architecture and docking standard.</p></div>

<div class="prc rv rv-d2"><div class="prc-d"></div><div class="text-[#28241f] text-[10px] font-mono mb-1">02</div><h3 class="text-[14px] font-semibold tracking-tight mb-0.5">Engineering</h3><p class="text-[#504a44] text-[12px] leading-[1.6]">Structural, thermal, and ECLSS design. FEA under all load cases. Trade studies for mass vs. reliability.</p></div>

<div class="prc rv rv-d3"><div class="prc-d"></div><div class="text-[#28241f] text-[10px] font-mono mb-1">03</div><h3 class="text-[14px] font-semibold tracking-tight mb-0.5">Simulation</h3><p class="text-[#504a44] text-[12px] leading-[1.6]">Build and run the digital twin. Verify system behaviour, failure modes, and long-duration performance.</p></div>

<div class="prc rv rv-d4"><div class="prc-d"></div><div class="text-[#28241f] text-[10px] font-mono mb-1">04</div><h3 class="text-[14px] font-semibold tracking-tight mb-0.5">Review & Deliver</h3><p class="text-[#504a44] text-[12px] leading-[1.6]">Independent review board assessment. Design package release — drawings, models, analysis reports, and simulation data.</p></div>

</div>

</div>

</section>

<!-- FAQ -->

<section id="faq" class="relative py-16 md:py-28 border-t" style="border-color:rgba(208,204,200,.03);background:rgba(6,6,8,.96)">

<div class="max-w-[1400px] mx-auto px-5 lg:px-9 lg:pl-[195px]">

07 — FAQ

<div class="hr-a h-px mb-8 rv rv-d1" style="background:rgba(208,204,200,.04)"></div>

<h2 class="text-[clamp(1.5rem,2.8rem,2.5rem)] font-bold tracking-[-.03em] leading-[.95] mb-8 rv rv-d1">Clarity up front</h2>

<div class="max-w-[560px] rv rv-d2">

<div class="fq-i"><div class="fq-q" onclick="toggleFaq(this)">What is a habitat module?<i data-lucide="plus" class="fq-ic w-3.5 h-3.5 text-[#28241f] flex-shrink-0"></i></div><div class="fq-a"><p class="text-[#504a44] text-[12px] leading-[1.6]">A pressurised cylinder — typically 4–7m in diameter and 6–12m long — that serves as the

basic building block of a space station or surface habitat. It contains structure, thermal control, life support, and interior fit-out for crew habitation or work.

How long does a programme take?

A single-module concept study: 6–8 weeks. A full station architecture with digital twin: 6–12 months. Lunar surface programmes with ISRU integration can extend to 18 months.

Do you manufacture hardware?

No. We deliver the design package — engineering drawings, 3D models, structural analysis, thermal models, and simulation environments. Manufacturing is handled by the prime contractor or their supply chain.

What standards do you work to?

NASA-STD-5009, ECSS-E-ST-32-08C, and IPCS where applicable. All structural analysis follows the NASA-STD-5002 fracture control approach. Materials are selected from the NASA materials database.

<!-- THE ARCHITECTURE -->

<section id="architecture" class="relative py-16 md:py-28 border-t" style="border-color:rgba(208,204,200,.03);background:rgba(6,6,8,.96)">

<div class="max-w-[1400px] mx-auto px-5 lg:px-9 lg:pl-[195px]">

08 — The Architecture

<div class="hr-a h-px mb-8 rv rv-d1" style="background:rgba(208,204,200,.04)"></div>

<h2 class="text-[clamp(1.5rem,2.8rem,2.5rem)] font-bold tracking-[.03em] leading-[.95] mb-3 rv rv-d1">How a habitat is put together</h2>

<p class="text-[#504a44] text-[12px] max-w-[340px] mb-10 rv rv-d2">Not a hunch — a system. Structure, thermal, life support, and simulation all working into each other.</p>

<div class="max-w-[700px] rv rv-d3">

<div class="pl-item"><div><h3 class="text-[14px] font-semibold tracking-tight mb-0.5">Structure</h3><p class="text-[#504a44] text-[12px] leading-[1.6]">Aluminium-lithium 2195

shell, machined ring frames, longeron stiffeners. All welds per AWS D17.1. Fracture control per NASA-STD-5002.

Thermal
Multi-layer insulation, fluid loop radiators, heat pipes for electronics. Sizing for worst-case solar exposure and full crew metabolic load.

ECLSS
Water recovery >98%, CO2 scrubbing via solid amine, trace contaminant control. Sabatier reactor for O2 recovery from CO2.

Digital Twin
Real-time co-simulation of structure, thermal, ECLSS, and power. Modelica-based with FMU export for partner integration.

<!-- ACTIVITY -->

<section id="activity" class="relative py-16 md:py-28 border-t" style="border-color:rgba(208,204,200,.03);background:rgba(6,6,8,.96)">

<div class="max-w-[1400px] mx-auto px-5 lg:px-9 lg:pl-[195px]">

09 — Activity

<div class="hr-a h-px mb-8 rv rv-d1" style="background:rgba(208,204,200,.04)"></div>

A year, *day by day*

Each cell is a working day. Dense weeks are engineering phases; quiet weeks are concept and review.

<div class="rv rv-d3 overflow-x-auto pb-2">

<div class="flex gap-[2px] min-w-[560px]" id="act-grid"></div>

<div class="flex justify-between mt-2 text-[8px] text-[#28241f] tracking-[.13em] uppercase min-w-[560px]">JanFebMarAprMayJunJulAugSepOctNovDec</div>

</div>

</div>

</section>

<!-- CONTACT -->

<section id="contact" class="relative py-16 md:py-28 border-t" style="border-color:rgba(208,204,200,.03)">

<div class="max-w-[1400px] mx-auto px-5 lg:px-9 lg:pl-[195px]">

<div class="grid grid-cols-1 lg:grid-cols-12 gap-12 lg:gap-6">

<div class="lg:col-span-5">

10 —
Project enquiry

<div class="hr-a h-px mb-8 rv rv-d1" style="background:rgba(208,204,200,.04)"></div>

<h2 class="text-[clamp(1.5rem,2.8rem,2.5rem)] font-bold tracking-[-.03em] leading-[-.95] mb-4
rv rv-d1">Ready to begin?</h2>

<p class="text-[#504a44] text-[12px] leading-[1.65] mb-7 rv rv-d2">Tell us about your
programme. We respond personally — with an honest scope and timeline, not a template.</p>

<div class="space-y-2.5 rv rv-d3">

<i data-lucide="mail" class="w-3 h-3"></i>ops@monolith.space

<div class="flex items-center gap-2 text-[12px] text-[#28241f]"><i data-lucide="map-pin" class="w-3 h-3"></i>Munich / Germany — Remote worldwide</div>

</div>

</div>

<div class="lg:col-span-5 lg:col-start-8 rv rv-d2">

<form id="cform" class="space-y-5" onsubmit="handleForm(event)">

<div><label class="block text-[8.5px] tracking-[-.16em] uppercase text-[#28241f] mb-1.5">Name</label><input type="text" required class="w-full bg-transparent border-b pb-2 text-[12px] text-[#d0ccc8] outline-none focus:border-[rgba(208,204,200,.12)] transition-colors placeholder:text-[#28241f]" style="border-color:rgba(208,204,200,.03)" placeholder="Your name"></div>

<div><label class="block text-[8.5px] tracking-[-.16em] uppercase text-[#28241f] mb-1.5">Email</label><input type="email" required class="w-full bg-transparent border-b pb-2 text-[12px] text-[#d0ccc8] outline-none focus:border-[rgba(208,204,200,.12)] transition-colors placeholder:text-[#28241f]" style="border-color:rgba(208,204,200,.03)" placeholder="you@agency.gov"></div>

<div><label class="block text-[8.5px] tracking-[-.16em] uppercase text-[#28241f] mb-1.5">Programme type</label>

```
<select required class="w-full bg-transparent border-b pb-2 text-[12px] text-[#504a44] outline-
none focus:border-[rgba(208,204,200,.12)] transition-colors appearance-none cursor-pointer"
style="border-color:rgba(208,204,200,.03)">
```

```
<option value="" class="bg-[#0a0908]">Select a type</option>
```

```
<option value="hab" class="bg-[#0a0908]">Habitat module</option>
```

```
<option value="surf" class="bg-[#0a0908]">Surface infrastructure</option>
```

```
<option value="orb" class="bg-[#0a0908]">Orbital station</option>
```

```
<option value="other" class="bg-[#0a0908]">Other</option>
```

```
</select>
```

```
</div>
```

```
<div><label class="block text-[8.5px] tracking-[.16em] uppercase text-[#28241f] mb-
1.5">Details</label><textarea rows="3" class="w-full bg-transparent border-b pb-2 text-[12px] text-
[#d0ccc8] outline-none focus:border-[rgba(208,204,200,.12)] transition-colors resize-none
placeholder:text-[#28241f]" style="border-color:rgba(208,204,200,.03)" placeholder="Brief
description of your programme"></textarea></div>
```

```
<div class="pt-1">
```

```
<button type="submit" class="group bg-[#d0ccc8] text-[#060608] text-[9px] tracking-[.1em]
uppercase font-medium px-5 py-2 rounded-full hover:bg-white transition-colors inline-flex items-
center gap-2"><span>Start a project enquiry</span><i data-lucide="arrow-right" class="w-3 h-3
group-hover:translate-x-0.5 transition-transform"></i></button>
```

```
<span class="block text-[#28241f] text-[9px] mt-2 tracking-wider">Takes about 60
seconds.</span>
```

```
</div>
```

```
</form>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- FOOTER -->
```

```
<footer id="footer" class="border-t py-12 md:py-14" style="border-
color:rgba(208,204,200,.03);background:rgba(6,6,8,.96)">
```

```
<div class="max-w-[1400px] mx-auto px-5 lg:px-9 lg:pl-[195px]">
```

```
<div class="grid grid-cols-2 md:grid-cols-4 gap-7 md:gap-5 mb-12">
```

<div class="col-span-2 md:col-span-1"><div class="text-[12px] font-bold tracking-[-.03em] mb-2.5">MONOLITH</div><p class="text-[#28241f] text-[11px] leading-[1.55]">Orbital architecture — habitat design, structural analysis, and digital twin simulation.</p></div>

<div><div class="text-[8.5px] tracking-[-.16em] uppercase text-[#28241f] mb-2.5">Navigation</div><div class="space-y-1">StartWhat we buildWorkProcessFAQ</div></div>

<div><div class="text-[8.5px] tracking-[-.16em] uppercase text-[#28241f] mb-2.5">Studio</div><div class="space-y-1">By the NumbersAll under one roofContact</div></div>

<div><div class="text-[8.5px] tracking-[-.16em] uppercase text-[#28241f] mb-2.5">Contact</div><div class="space-y-1">ops@monolith.spacemonolith.spaceRemote — Germany & worldwide</div></div>

<div class="mt-4"><div class="text-[8.5px] tracking-[-.16em] uppercase text-[#28241f] mb-2.5">Legal</div><div class="space-y-1">ImpressumPrivacyCookies</div></div>

</div>

</div>

<div class="border-t pt-6 flex flex-col md:flex-row md:items-center md:justify-between gap-2.5" style="border-color:rgba(208,204,200,.03)">

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<button onclick="document.getElementById('cookie').classList.add('show')" class="text-[#28241f] hover:text-[#504a44] transition-colors text-[9px] tracking-wider">Cookie settings</button>

</div>

</div>

</footer>

</main>

```

<div id="cookie"><div class="max-w-[1400px] mx-auto px-4 lg:px-9 lg:pl-[195px] pb-4"><div
class="bg-[#0a0908]/95 backdrop-blur-xl border rounded-sm p-4" style="border-
color:rgba(208,204,200,.03)"><div class="flex flex-col md:flex-row md:items-center gap-3.5"><div
class="flex-1"><div class="text-[11px] font-medium mb-0.5">Cookies & technologies</div><p
class="text-[#504a44] text-[10px] leading-relaxed">This site uses technically necessary functions.
Beyond those, we would like to measure anonymously how the site is used — only with your
consent.</p></div><div class="flex items-center gap-1.5 flex-shrink-0"><button
onclick="document.getElementById('cookie').classList.remove('show')" class="text-[9px] tracking-
wider uppercase border px-3 py-1.5 rounded-full hover:bg-white/[.015] transition-colors"
style="border-color:rgba(208,204,200,.06)">Reject all</button><button
onclick="document.getElementById('cookie').classList.remove('show')" class="text-[9px] tracking-
wider uppercase bg-[#d0ccc8] text-[#060608] px-3 py-1.5 rounded-full hover:bg-white transition-
colors">Accept all</button><button class="text-[#504a44] hover:text-[#d0ccc8] transition-colors
text-[9px] px-1">Settings</button></div></div></div></div>

```

```

<div id="toast"><i data-lucide="check-circle" class="w-3.5 h-3.5 text-[#D4623B]"></i><span
class="text-[11px]">Enquiry sent — we'll be in touch.</span></div>

```

```

<script>

```

```

lucide.createIcons();

```

```

gsap.registerPlugin(ScrollTrigger);

```

```

const isMobile=window.innerWidth<768;

```

```

const

```

```

secs=['start','numbers','build','engineer','work','studio','process','faq','architecture','activity','contact',
'footer'];

```

```

let activeSec='start',curScene=0;

```

```

// ===== LOADER =====

```

```

let lp=0;const lf=document.getElementById('ld-f'),ln=document.getElementById('ld-n');

```

```

const

```

```

li=setInterval(()=>{lp+=Math.random()*14+6;if(lp>=100){lp=100;clearInterval(li);setTimeout(finishLoa
d,280)}lf.style.width=lp+'%';ln.textContent=String(Math.round(lp)).padStart(3,'0')},105);

```

```

function

```

```

finishLoad(){document.getElementById('loader').classList.add('out');document.querySelectorAll('.hl
span').forEach((s,i)=>setTimeout(()=>s.classList.add('in'),60+i*110));document.querySelectorAll('#h-
tag,#h-sub,#h-cta').forEach((e,i)=>setTimeout(()=>e.classList.add('in'),380+i*90))}

```

```

// ===== THREE.JS =====

const canvas=document.getElementById('c3d');

const renderer=new THREE.WebGLRenderer({canvas,antialias:!isMobile,alpha:true});

renderer.setPixelRatio(Math.min(window.devicePixelRatio,isMobile?1.5:2));

renderer.setSize(window.innerWidth>window.innerHeight);

renderer.toneMapping=THREE.ACESFilmicToneMapping;

renderer.toneMappingExposure=1.0;

renderer.shadowMap.enabled=!isMobile;


const scene=new THREE.Scene();

scene.fog=new THREE.FogExp2(0x060608,0.045);

const camera=new THREE.PerspectiveCamera(36>window.innerWidth>window.innerHeight,0.1,100);

camera.position.set(0,0.2,5.8);


// Lights

const ambLight=new THREE.AmbientLight(0x556677,0.3);scene.add(ambLight);

const sunLight=new THREE.DirectionalLight(0xffff00,1.2);sunLight.position.set(5,7,4);

if(!isMobile){sunLight.castShadow=true;sunLight.shadow.mapSize.set(1024,1024)}scene.add(sunLight);

const fillLight=new THREE.DirectionalLight(0x445566,0.2);fillLight.position.set(-4,2,-3);scene.add(fillLight);

const rimLight=new THREE.PointLight(0xD4623B,0.12,10);rimLight.position.set(-3,1,2);scene.add(rimLight);


// ===== CREATE MAIN MODEL (REPLACEABLE) =====

// To replace: modify this function. Return a THREE.Group.

// The group will be positioned/scaled/rotated by the animation system.

// All material references are stored in group.userData for scene switching.

function createMainModel(){

const g=new THREE.Group();

const hM=new THREE.MeshStandardMaterial({color:0x888580,metalness:0.6,roughness:0.28});

const dM=new THREE.MeshStandardMaterial({color:0x3a3835,metalness:0.5,roughness:0.4});

```

```

const gM=new
THREE.MeshStandardMaterial({color:0x667788,metalness:0.3,roughness:0.05,transparent:true,opacity:0.22,side:THREE.DoubleSide});

const aM=new
THREE.MeshStandardMaterial({color:0xD4623B,metalness:0.4,roughness:0.3,emissive:0xD4623B,emissiveIntensity:0.1});

const sM=new
THREE.MeshStandardMaterial({color:0x151d30,metalness:0.3,roughness:0.2,emissive:0x0a1120,emissiveIntensity:0.08});

const gndM=new THREE.MeshStandardMaterial({color:0x0a0908,roughness:1,metalness:0});

const seg=isMobile?16:28;

// Ground + grid

const gnd=new THREE.Mesh(new THREE.PlaneGeometry(5,5),gndM);gnd.rotation.x=-
Math.PI/2;gnd.position.y=-0.012;if(!isMobile)gnd.receiveShadow=true;g.add(gnd);

const gr=new THREE.GridHelper(5,20,0x181614,0x121010);gr.position.y=-0.005;g.add(gr);

// Main hull (horizontal cylinder)

const cyl=new THREE.Mesh(new
THREE.CylinderGeometry(Math.max(.01,.65),Math.max(.01,.65),2.0,seg,1,true),hM);

cyl.rotation.z=Math.PI/2;if(!isMobile)cyl.castShadow=true;g.add(cyl);

// End domes

const cG=new
THREE.SphereGeometry(Math.max(.01,.65),seg,Math.max(4,Math.floor(seg/2)),0,Math.PI*2,0,Math.
PI/2);

const c1=new THREE.Mesh(cG,hM);c1.position.x=1.0;c1.rotation.z=-Math.PI/2;g.add(c1);

const c2=new THREE.Mesh(cG,hM);c2.position.x=-1.0;c2.rotation.z=Math.PI/2;g.add(c2);

// Viewport windows

for(let i=0;i<(isMobile?4:6);i++){

const a=(i/(isMobile?4:6))*Math.PI*2;

const w=new THREE.Mesh(new THREE.PlaneGeometry(.12,.25),gM);

```



```
w.position.set(0,Math.sin(a)*.67,Math.cos(a)*.67);w.lookAt(0,Math.sin(a)*2,Math.cos(a)*2);g.add(w);  
}
```

```
// Solar truss
```

```
const tr=new THREE.Mesh(new THREE.BoxGeometry(.025,.025,2.6),dM);tr.position.y=.9;g.add(tr);
```

```
// Solar panels
```

```
const pG=new THREE.BoxGeometry(.5,.006,1.0);
```

```
for(let i=-1;i<=1;i++){const p=new THREE.Mesh(pG,sM);p.position.set(i*.55,.9,0);g.add(p)}
```

```
// Docking port + accent ring
```

```
const dk=new THREE.Mesh(new  
THREE.CylinderGeometry(Math.max(.01,.1),Math.max(.01,.1),.12,12),dM);dk.position.x=1.12;dk.rotation.z=Math.PI/2;g.add(dk);
```

```
const dr=new THREE.Mesh(new  
THREE.TorusGeometry(Math.max(.01,.12),.012,8,20),aM);dr.position.x=1.2;dr.rotation.y=Math.PI/2;g.add(dr);
```

```
// Rear radiator panels
```

```
const rG=new THREE.BoxGeometry(.004,.7,.35);
```

```
for(let i=-1;i<=1;i+=2){const r=new THREE.Mesh(rG,dM);r.position.set(-1.2,i*.45,0);g.add(r)}
```

```
// Communication dish
```

```
const dsh=new THREE.Mesh(new  
THREE.SphereGeometry(Math.max(.01,.13),12,8,0,Math.PI*2,0,Math.PI/3),hM);dsh.position.set(-.25,1.15,0);dsh.rotation.x=Math.PI;g.add(dsh);
```

```
// Accent position lights
```

```
const lG=new THREE.SphereGeometry(Math.max(.001,.012),6,6);
```

```
[[.7,0,0],[-.7,0,0],[0,.65,0],[0,-.65,0],[.9,.25,0],[.9,-.25,0],[-.9,.25,0],[-.9,-.25,0]].forEach(p=>{
```

```
const l=new THREE.Mesh(lG,aM);l.position.set(p[0],p[1],p[2]);g.add(l);
```

```
});
```

```

// Hull ring bands
for(let i=-2;i<=2;i++){
  if(!i)continue;

  const rn=new THREE.Mesh(new
THREE.TorusGeometry(Math.max(.01,.67),.005,6,isMobile?20:40),dM);

  rn.position.x=i*.32;rn.rotation.y=Math.PI/2;g.add(rn);
}

// Thruster housing (rear)
const thG=new THREE.CylinderGeometry(Math.max(.01,.18),Math.max(.01,.22),.15,12);

const th=new THREE.Mesh(thG,dM);th.position.x=-
1.18;th.rotation.z=Math.PI/2;if(!isMobile)th.castShadow=true;g.add(th);

const thR=new THREE.Mesh(new
THREE.TorusGeometry(Math.max(.01,.23),.008,8,16),dM);thR.position.x=-
1.28;thR.rotation.y=Math.PI/2;g.add(thR);

g.userData={hullMat:hM,darkMat:dM,glassMat:gM,accentMat:aM,solarMat:sM};
return g;
}

const model=createMainModel();
model.position.set(2.0,-.15,0);
scene.add(model);

// ===== ANIMATION STATE =====
let cur={x:2,y:-.15,z:0,s:1,rm:1},tgt={...cur};
let mx=0,my=0,smx=0,smy=0,scrollProg=0;

document.addEventListener('mousemove',e=>{mx=(e.clientX/window.innerWidth-
.5)*2;my=(e.clientY/window.innerHeight-.5)*2});

window.addEventListener('scroll',()=>{scrollProg=window.pageYOffset/(document.body.scrollHeight-
window.innerHeight)});

```

```

const SEC_POS={
  start:{x:2,y:-.15,z:0,s:1,rm:1},
  numbers:{x:2.8,y:-.2,z:-.5,s:.82,rm:.65},
  build:{x:3.3,y:-.25,z:-1.3,s:.65,rm:.45},
  engineer:{x:3.8,y:-.3,z:-2.2,s:.5,rm:.3},
  work:{x:0,y:-1.8,z:-5,s:.22,rm:.1},
  studio:{x:-2.2,y:-.2,z:-3.2,s:.35,rm:.18},
  process:{x:-2.8,y:0,z:-3.8,s:.3,rm:.15},
  faq:{x:0,y:-1.2,z:-5.5,s:.16,rm:.08},
  architecture:{x:0,y:0,z:-6.5,s:.12,rm:.05},
  activity:{x:0,y:0,z:-7.5,s:.09,rm:.03},
  contact:{x:0,y:0,z:-8.5,s:.07,rm:.02},
  footer:{x:0,y:0,z:-9.5,s:.05,rm:.01}
};

function setTarget(id){const p=SEC_POS[id];if(p){tgt.x=p.x;tgt.y=p.y;tgt.z=p.z;tgt.s=p.s;tgt.rm=p.rm}}

const SCENES=[

  {name:'ORBIT',hull:0x888580,dark:0x3a3835,glass:0x667788,sun:0xffff0dd,sunl:1.2,amb:0x556677,amb:0x556677,amb:0x556677,rim:0xD4623B,riml:.12},

  {name:'LUNAR',hull:0x908880,dark:0x454038,glass:0x778877,sun:0xffeedd,sunl:1.4,amb:0x665544,amb:0x665544,amb:0x665544,rim:0xD4623B,riml:.15},

  {name:'DEEP
SPACE',hull:0x505058,dark:0x2a2a30,glass:0x334455,sun:0x445566,sunl:.5,amb:0x223344,amb:0x223344,amb:0x223344,rim:0xD4623B,riml:.35}

];

function changeScene(idx){
  curScene=idx;const s=SCENES[idx];document.getElementById('sc-name').textContent=s.name;

```

```
sunLight.color.set(s.sun);sunLight.intensity=s.sunI;ambLight.color.set(s.amb);ambLight.intensity=s.ambI;
```

```
scene.fog.color.set(s.fog);renderer.setClearColor(s.fog,0);rimLight.intensity=s.rimI;  
}
```

```
// ===== RENDER LOOP =====
```

```
const clock=THREE.Clock();
```

```
function animate(){
```

```
requestAnimationFrame(animate);const t=clock.getElapsedTime();
```

```
smx+=(mx-smx)*.018;smy+=(my-smy)*.018;
```

```
cur.x+=(tgt.x-cur.x)*.02;cur.y+=(tgt.y-cur.y)*.02;cur.z+=(tgt.z-cur.z)*.02;
```

```
cur.s+=(tgt.s-cur.s)*.02;cur.rm+=(tgt.rm-cur.rm)*.02;
```

```
model.position.set(cur.x+smx*.1*cur.s,cur.y-smy*.07*cur.s,cur.z);
```

```
model.scale.setScalar(cur.s);
```

```
model.rotation.y=Math.sin(t*.06*cur.rm)*.04+scrollProg*.35;
```

```
camera.position.x+=(smx*.08-camera.position.x)*.01;
```

```
camera.position.y+=(.2-smy*.05-camera.position.y)*.01;
```

```
camera.lookAt(cur.x*.2,cur.y*.2,cur.z);
```

```
const sp=SCENES[curScene];const ud=model.userData;
```

```
ud.hullMat.color.lerp(new THREE.Color(sp.hull),.02);
```

```
ud.darkMat.color.lerp(new THREE.Color(sp.dark),.02);
```

```
ud.glassMat.color.lerp(new THREE.Color(sp.glass),.02);
```

```
renderer.render(scene,camera);
```

```
}
```

```
animate();
```

```
window.addEventListener('resize',()=>{camera.aspect=window.innerWidth/window.innerHeight;camera.updateProjectionMatrix();renderer.setSize(window.innerWidth>window.innerHeight)});
```

```
// ===== NAV =====
```

```
const topnav=document.getElementById('topnav');let lastSY=0;
```

```

window.addEventListener('scroll',()=>{const
y=window.pageYOffset;if(y>35)topnav.classList.add('scrolled');else
topnav.classList.remove('scrolled');if(y>lastSY&& y>280&&!mobOpen)topnav.classList.add('hide');else
topnav.classList.remove('hide');lastSY=y});

```

```

// ===== MOBILE MENU =====

```

```

const
menubtn=document.getElementById('menubtn'),mobmenu=document.getElementById('mobmenu')
;

```

```

const
ml1=document.getElementById('ml1'),ml2=document.getElementById('ml2'),ml3=document.getEle
mentById('ml3');

```

```

let mobOpen=false;

```

```

function
toggleMob(){mobOpen=!mobOpen;mobmenu.classList.toggle('open');if(mobOpen){ml1.style.transfo
rm='rotate(45deg) translate(2px,2px)';ml2.style.opacity='0';ml3.style.transform='rotate(-45deg)
translate(2px,-
2px)';document.body.style.overflow='hidden'}else{ml1.style.transform='none';ml2.style.opacity='1';
ml3.style.transform='none';document.body.style.overflow=""}}

```

```

menubtn.addEventListener('click',toggleMob);

```

```

document.querySelectorAll('.ml').forEach(l=>l.addEventListener('click',()=>{if(mobOpen)toggleMob()
}));

```

```

// ===== SIDEBAR =====

```

```

const sbLinks=document.querySelectorAll('#sb-links .sb-link');

```

```

const secObs=new
IntersectionObserver(entries=>{entries.forEach(e=>{if(e.isIntersecting){activeSec=e.target.id;setTarge
t(activeSec);sbLinks.forEach(l=>l.classList.toggle('active',l.dataset.target===activeSec)}})},{rootMargi
n:'-28% 0px -65% 0px'});

```

```

secs.forEach(id=>{const el=document.getElementById(id);if(el)secObs.observe(el)});

```

```

sbLinks.forEach(l=>l.addEventListener('click',()=>{const
t=document.getElementById(l.dataset.target);if(t)t.scrollIntoView({behavior:'smooth'}}));

```

```

// Keyboard

```

```

document.addEventListener('keydown',e=>{if(e.key==='ArrowUp' || e.key==='w' || e.key==='W'){e.pre
ventDefault();changeScene((curScene+2)%3)}if(e.key==='ArrowDown' || e.key==='s' || e.key==='S'){e.p
reventDefault();changeScene((curScene+1)%3)}if(e.key==='k' || e.key==='K')document.getElemenByI
d('contact').scrollIntoView({behavior:'smooth'}}));

```

```
// ===== REVEAL =====
```

```
const rvObs=new  
IntersectionObserver(entries=>{entries.forEach(e=>{if(e.isIntersecting)e.target.classList.add('in')})),{threshold:.04,rootMargin:'0px 0px -20px 0px'}};  
  
document.querySelectorAll('.rv,.hr-a').forEach(el=>rvObs.observe(el));
```

```
// ===== STAT COUNTERS =====
```

```
const svObs=new  
IntersectionObserver(entries=>{entries.forEach(e=>{if(!e.isIntersecting)return;const  
el=e.target,tgt=parseInt(el.dataset.t),sfx=el.dataset.s|'';let c=0;const steps=38,inc=tgt/steps;const  
timer=setInterval(()=>{c+=inc;if(c>=tgt){c=tgt;clearInterval(timer)}el.textContent=Math.round(c)+sfx,  
24});svObs.unobserve(el)})),{threshold:.5}};  
  
document.querySelectorAll('.sv').forEach(el=>svObs.observe(el));
```

```
// ===== FAQ =====
```

```
function toggleFaq(q){const a=q.nextElementSibling,ic=q.querySelector('.fq-  
ic'),isOpen=a.classList.contains('open');document.querySelectorAll('.fq-  
a').forEach(f=>f.classList.remove('open'));document.querySelectorAll('.fq-  
ic').forEach(i=>i.style.transform='rotate(0deg)');if(!isOpen){a.classList.add('open');ic.style.transform='  
rotate(45deg)'}}
```

```
// ===== ACTIVITY GRID =====
```

```
(function(){const grid=document.getElementById('act-grid'),seed=41;function sr(s){let  
x=Math.sin(s)*10000;return x-Math.floor(x)}for(let w=0;w<52;w++)for(let d=0;d<7;d++){const  
cell=document.createElement('div');cell.className='ac';const r=sr(seed+w*7+d);if(r>.4){const  
lv=r>.92?5:r>.76?4:r>.6?3:r>.49?2:1;cell.classList.add('l'+lv)}grid.appendChild(cell)}}();
```

```
// ===== FORM =====
```

```
function handleForm(e){e.preventDefault();const  
t=document.getElementById('toast');t.classList.add('show');e.target.reset();setTimeout(()=>t.classList  
.remove('show'),3000)}
```

```
// ===== GSAP PARALLAX =====
```

```
if(!isMobile){gsap.utils.toArray('.dem img').forEach(img=>{gsap.fromTo(img,{y:-  
12},{y:12,ease:'none',scrollTrigger:{trigger:img.closest('.dem'),start:'top bottom',end:'bottom  
top',scrub:1}})});}
```

</script>

</body>

</html>